Database Management Systems

Lab-2: E-R Diagram

Lab Objectives

- Develop an E-R Diagram using a standard modeling too.
- Translate the E-R model into a relational database schema in MySQL.
- Use SQL DDL statements to create tables with appropriate data types and constraints.
- Insert and retrieve data using DML commands (INSERT, SELECT).

Lab Submission Instructions(10%)

- Write your SQL code using a standard editor such as VS Code, and save the file with a .sql extension.
- Verify that all queries execute successfully in your MySQL environment before submission.
- Complete the lab work in the same project groups previously assigned for your DBMS project.
- Submit your .sql file at the end of this lab before 11:59 AM.
- Submit a separate file for your ER Diagram: either export it as an .xml file or save it as ”filename”.drawio.
- Note: Identical or copied submissions across different groups will receive zero marks.

Develop an Entity-Relationship Diagram based on the company case study from our previous lessons.
The diagram must be created using a standard Entity-Relationship drawing tool. I recommend using the online tool draw.io; (Link)

The diagram must include the following details:

- Company organized into DEPARTMENT. Each department has unique name and a particular employee who manages the department. Start date for the manager is recorded.
- Department may have several locations.
- A department controls a number of PROJECT. Projects have a unique name, number and a single location.

- Company's EMPLOYEE name, ssno, address, salary, sex and birth date are recorded. An employee is assigned to one department, but may work for several projects. Number of hours/week an employee works on each project is recorded; also keep track of the direct supervisor of each employee.
- Employee's DEPENDENT are tracked for health insurance purposes (dependent name, birthdate, relationship to employee).

1. Create a database Named "company" in your MySQL environment
2. Create a table for each entity (EMPLOYEE, DEPARTMENT, PROJECT, and DEPENDENT). Define appropriate data types and constraints for each attribute.
3. Show the schema for all tables in the database.
4. Insert five sample records into each table and execute a query to display all inserted data.
5. Add a new attribute named "Project_Title" to the DEPENDENT entity. Modify the corresponding table and then display the updated table structure.